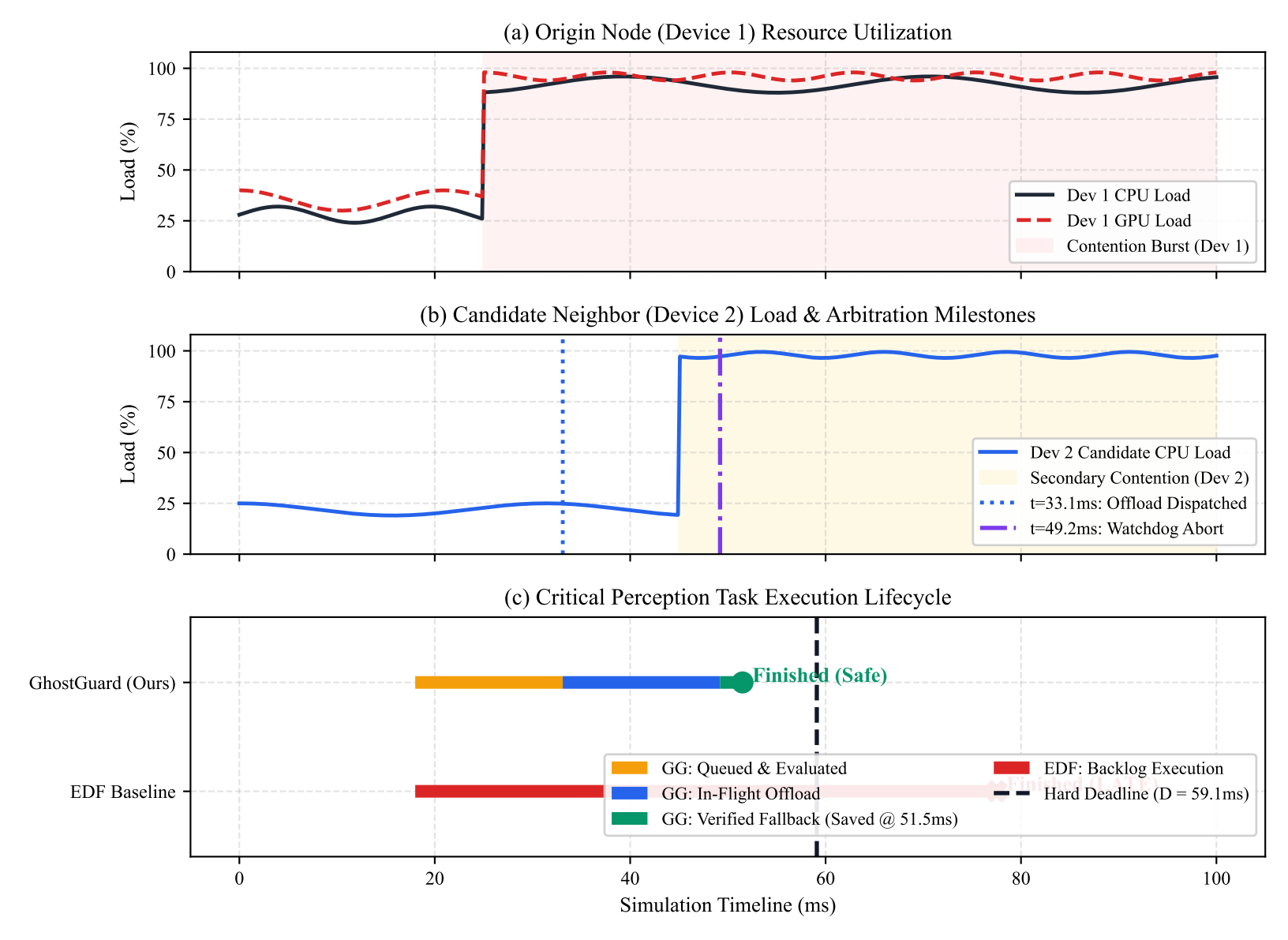
(a) Origin Node (Device 1) Resource Utilization
Load (%)
100
75
50
25
0
Dev 1 CPU Load
Dev 1 GPU Load
Contention Burst (Dev 1)
(b) Candidate Neighbor (Device 2) Load & Arbitration Milestones
Load (%)
100
75
50
25
0
Dev 2 Candidate CPU Load
Secondary Contention (Dev 2)
t=33.1ms: Offload Dispatched
t=49.2ms: Watchdog Abort
(c) Critical Perception Task Execution Lifecycle
GhostGuard (Ours)
EDF Baseline
Finished (Safe)
GG: Queued & Evaluated
GG: In-Flight Offload
GG: Verified Fallback (Saved @ 51.5ms)
EDF: Backlog Execution
Hard Deadline (D = 59.1ms)
0
20
40
60
80
100
Simulation Timeline (ms)